SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – PROMPT ENGINEERING CHALLENGE

Course Code:	CSA65	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL2, BL3)	**Assessment:**	Assessment Tool 1 – Transformer Mechanics Worksheet
Weightage:	40% of CIA	**Total Marks:**	2 * 50 = 100 (2 Questions)
CO1 Statement:	*apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)*		
Student Name:	Y. Jayesh Ranjan	**Reg. No.:**	192472201
Section / Batch:	CSE – AI	**Date:**	02/08/26

Instructions to Students
• Answer 2 questionss.Each questions carries 50 marks..Total100 Marks.
1. Design appropriate prompts for the given tasks using suitable prompt engineering techniques.
2. Select and apply the appropriate prompting strategy (Zero-shot, One-shot, Few-shot, Chain-of-Thought, or Structured Prompting) based on the given scenario.
3. Include clear instructions, relevant context, necessary constraints, and the expected output format while designing each prompt.
4. Evaluate and refine the designed prompts to improve their clarity, effectiveness, and quality by applying prompt engineering best practices.
5. Design prompts for a variety of real-world applications, such as text generation, summarization, translation, question answering, code generation, and conversational AI.
6. Justify the designed prompt by briefly explaining the selected prompting technique and how it helps achieve the desired output.

Code of Conduct
I Y. Jayesh Ranjan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jayesh Ranjan

SECTION A – PROMPT ENGINEERING CHALLENGE

5. Create a Structured Prompt to generate a professional email requesting leave from work.

Create a Structured Prompt to Generate a Professional Email Requesting Leave from Work

1. Prompting Strategy Selected

Technique: Structured Prompting

Reason for Selection

Structured Prompting is suitable because generating a professional email requires the AI to follow a fixed structure, maintain a formal tone, include specific information, and produce the output in a predefined format. This reduces ambiguity and improves consistency.

2. Initial Prompt

Generate a professional email requesting leave from work.

Context:

You are an employee writing to your manager requesting leave.

Instructions:

- Mention the reason for leave.
- Include the leave dates.
- Use a polite and formal tone.
- Request approval.
- End the email professionally.

Output Format:

Subject:

Email:

3. Evaluation of the Initial Prompt

Strengths

- Provides a clear task.
- Includes the required output format.
- Specifies a formal tone.

Limitations

- Does not specify the employee's role.
- No word limit is provided.
- The leave reason is not clearly defined.
- Does not mention handing over pending work.
- Lacks detailed formatting instructions.

4. Refined Structured Prompt (Improved Prompt)

Role:

You are a professional software engineer working in a corporate company.

Task:

Write a formal email requesting leave from your reporting manager.

Context:

You need leave from 15 August 2026 to 16 August 2026 to attend an important family function.

Instructions:

1. Write a clear and professional subject line.
2. Begin with a formal greeting.
3. Mention the purpose of the email.
4. Clearly state the leave dates.
5. Explain the leave reason briefly.
6. Mention that all current tasks are completed or handed over to teammates.
7. Politely request leave approval.
8. Express gratitude to the manager.
9. End with a professional closing.
10. Keep the email between 120 and 150 words.
11. Use simple, professional English without grammatical mistakes.

Expected Output Format:

Subject:

Email:

Regards,

Employee Name

Software Engineer

5. Expected Output

Subject: Leave Request for 15–16 August 2026

Email:

Dear Manager,

I hope you are doing well.

I would like to request leave for two days, from 15 August 2026 to 16 August 2026, as I need to attend an important family function.

I have completed all my assigned tasks and shared the remaining work with my teammates to ensure that there is no disruption during my absence.

I kindly request you to approve my leave application. I will resume work on 17 August 2026 and continue my responsibilities as usual.

Thank you for your understanding and support.

Regards,

Employee Name

Software Engineer

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6. Real-World Application

This prompt can be used in:

- HR Management Systems
- Corporate Email Assistants
- AI Writing Assistants
- Employee Self-Service Portals
- Business Communication Tools

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7. Justification of the Prompting Technique

Structured Prompting was selected because it clearly defines:

- Role
- Task
- Context
- Instructions
- Constraints
- Expected Output Format

This helps the AI generate accurate, professional, and consistent emails while minimizing ambiguity.

26. Design a prompt that explains a programming error and suggests possible fixes.

Design a Prompt that Explains a Programming Error and Suggests Possible Fixes

1. Prompting Strategy Selected

Technique: Chain-of-Thought Prompting

Reason for Selection

Programming errors require logical reasoning to determine the error type, identify the root cause, suggest solutions, and explain the corrected code. Chain-of-Thought Prompting guides the AI through these reasoning steps, producing clear and educational explanations.

2. Initial Prompt

Explain the following programming error.

Include:

- Error type
- Cause
- Explanation
- Possible fixes
- Corrected code

3. Evaluation of the Initial Prompt

Strengths

- Simple and understandable.
- Requests the basic information.

Limitations

- Does not specify the programming language.
- No structured reasoning process.
- Does not request beginner-friendly explanations.
- Missing prevention tips.
- Output format is not defined.

4. Refined Prompt (Improved Prompt)

Role:

You are an experienced programming instructor and software developer.

Task:

Analyze the programming error provided by the user and explain it in a beginner-friendly manner.

Context:

The user will provide:

- Programming language
- Error message
- Code snippet

Instructions:

1. Identify the programming language.
2. Identify the error type.

3. Explain the error step by step.
4. Mention why the error occurred.
5. Identify the probable line causing the error.
6. Suggest at least three possible fixes.
7. Provide corrected code.
8. Explain how the corrected code solves the problem.
9. Give practical tips to avoid similar errors in future.
10. Use simple English suitable for beginners.

Expected Output Format:

Programming Language:

Error Type:

Cause:

Step-by-Step Explanation:

Possible Fixes:
1.
2.
3.

Corrected Code:

How the Correction Works:

Prevention Tips:

5. Example Expected Output

Programming Language

Python

Error Type

SyntaxError

Cause

A closing parenthesis is missing in the print() statement.

Step-by-Step Explanation

- Python reads the code from left to right.
- The print() function requires matching opening and closing parentheses.
- Since one parenthesis is missing, Python cannot understand where the statement ends.
- Therefore, it raises a SyntaxError before executing the program.

Possible Fixes

1. Add the missing closing parenthesis.
2. Check all brackets and quotation marks.
3. Use an IDE that highlights syntax errors automatically.

Corrected Code

```
print("Hello World")
```

How the Correction Works

The corrected code includes the required closing parenthesis, making the statement syntactically correct. Python can now successfully execute the program and display the output.

Prevention Tips

- Use a code editor with syntax highlighting.
- Match every opening bracket with a closing bracket.
- Test your code after every small change.
- Format your code regularly.
- Read the error message carefully before making corrections.

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6. Real-World Application

This prompt can be used in:

- AI Coding Assistants
- Online Programming Tutors
- Software Development Tools
- Educational Coding Platforms
- Technical Interview Preparation
- IDE-based AI Debugging Systems

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7. Justification of the Prompting Technique

Chain-of-Thought Prompting was selected because debugging requires systematic reasoning. The prompt guides the AI to:

- Identify the error
- Analyze the cause
- Explain the problem logically
- Suggest multiple solutions
- Generate corrected code
- Provide prevention tips

This approach improves the clarity, accuracy, and educational value of the response, making it especially useful for beginners and developers learning to debug programs.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technique Selection	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Output Effectiveness	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Prompt Craftsmanship	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Prompt-Output Documentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Design Rationale	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________